Which View Should Teach? Label-Aware Flip Consistency for Generalizable Polyp Segmentation

Authors

***Abstract*—Polyp segmentation networks lose accuracy on centres unseen in training. A mirrored polyp is still a polyp, so a reliable model should predict the mirrored mask for a mirrored image; we observe that trained models do not, and are least flip-consistent on the hardest test sets. Flip consistency regularization is a natural remedy, but it comes from semi-supervised learning, where no label says which view is correct. We show that, in binary segmentation with labels available, a cross-entropy consistency pull toward a teacher view is a reweighted supervised pull: it moves the student view toward the label only where the teacher is closer to it. Building on this identity, we propose label-aware flip-consistency training (FCT), in which each view is pulled at full strength only toward a more accurate view, and never toward a misclassification it has avoided. The rule's one constant is set by the decision threshold rather than tuned, and FCT leaves the network and its inference unchanged. Experiments with three representative architectures on five public benchmarks, each repeated over six random seeds, show that FCT improves Dice on the unseen ETIS dataset for all three networks, and the five-set mean on every seed for Polyp-PVT and PraNet-V2, whose flip inconsistency it also markedly reduces. Notably, FCT improves PraNet-V2 on ETIS by 3.6 Dice points.**

***Index Terms*—Polyp segmentation, domain generalization, consistency regularization, flip equivariance, self-distillation.**

I. Introduction

Polyp segmentation networks that are accurate on their training data lose much of that accuracy at new centres, whose images differ in endoscopes, illumination and patients [1], [2]. This gap limits their clinical value: colorectal cancer is among the leading causes of cancer death [3], and automatic segmentation aims to help endoscopists, who miss about one in five polyps [4]. Recent networks reach high Dice on test images from their training datasets [5]–[9], but much lower Dice elsewhere. In our experiments, for example, Polyp-PVT [7] reaches 93.2% mean Dice on CVC-ClinicDB but only 78.6% on ETIS, a dataset from another hospital. For clinical use, closing this gap matters more than further gains on the training distribution.

Polyp segmentation also has a symmetry that standard training does not enforce. A mirrored polyp is still a polyp, so the correct mask of a mirrored image is the mirrored mask. We measure departures from this flip equivariance with the flip gap: one minus the Dice overlap between the binary masks predicted for an image and for its flipped copy, once flipped back, averaged over horizontal and vertical flips. As Fig. 1 shows, the flip gap of trained models is far from zero, and it is largest on the two hardest test sets, CVC-ColonDB and ETIS. We hypothesize that a model whose answer changes with a mirror image relies partly on cues that do not belong to the polyp, and that such cues are unlikely to transfer to new centres.

Two remedies are common. Flip augmentation trains on mirrored copies with mirrored labels, but each copy is constrained only by its own label, never by the prediction on its mirror. Consistency regularization adds that constraint by penalizing disagreement between the predictions on the views. It comes from semi-supervised learning, where agreement between views is the only signal that unlabeled images provide [10]–[12], and medical variants enforce it under spatial transformations [13], [14]. Since no label says which view is right, the views are treated as equally reliable or weighed by the model's own confidence. Supervised uses of consistency, such as AugMix [15] and R-Drop [16], keep this symmetry even though the label is known, so a correct view can be pulled toward a wrong one.

We show that, under full supervision, the label settles which view should teach. For binary segmentation with a cross-entropy consistency loss toward a detached teacher view, the gradient on the student equals the supervised gradient scaled by one minus the ratio of the teacher's error to the student's. Similar decompositions of the distillation gradient are known for classification [17], [18]; here we apply the identity pixel by pixel, across views of one image. A consistency pull is therefore a reweighted supervised pull: it moves the student toward the label where the teacher is closer to it, and away from the label where the student is closer. Unweighted consistency mixes both kinds of pulls in every image, and only the label can tell them apart.

Building on this identity, we propose label-aware flip-consistency training (FCT). Each image is presented upright, flipped horizontally and flipped vertically, and all three views are supervised. Each flipped view is pulled toward the upright view at full strength wherever it lags behind it. Where it leads, the pull is kept only if the upright view is correct, scaled by how far the upright prediction lies on the correct side of the 0.5 threshold; this weak pull keeps the views consistent without moving either across the threshold, much as label smoothing tempers overconfident targets [19]. The upright view, in turn, learns from the flipped view closest to the label wherever that view is more accurate. The rule's single constant is set by the decision threshold rather than tuned, and FCT changes only the training objective: it multiplies the training time by about 3.5 and adds nothing at inference. To the best of our knowledge, FCT is the first rule that uses the label to decide, pixel by pixel and across views of one network, which view teaches.

We add FCT, without retuning, to three segmentation networks of different design: Polyp-PVT [7], built on a pyramid vision transformer; PraNet-V2 [8], a convolutional network with dual-supervised reverse attention; and ASGNet [9], which guides its features with spectral cues and already trains with

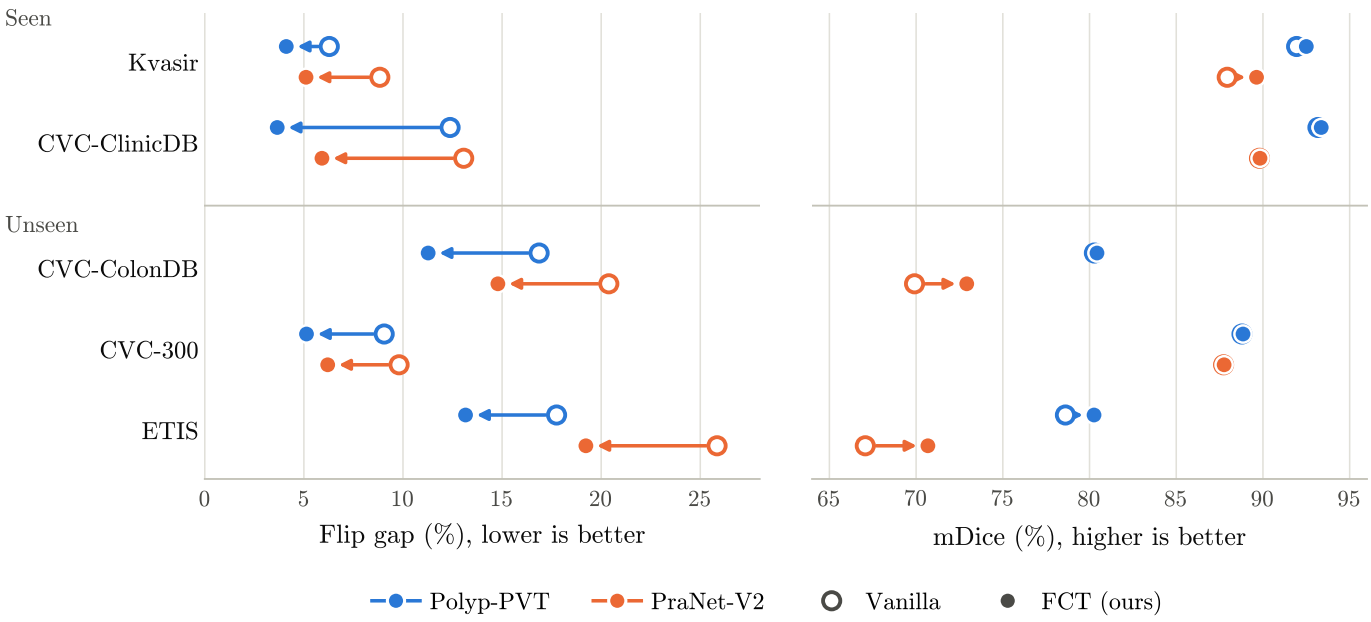


Fig. 1. Flip gap (left; one minus the Dice overlap between the binary masks predicted for an image and for its flipped copy, once flipped back, averaged over horizontal and vertical flips) and mDice (right) on five test sets, without (hollow) and with (filled) FCT. Both networks are least flip-consistent on CVC-ColonDB and ETIS, the two sets where they are least accurate. FCT lowers the flip gap on every set, and its largest mDice gain, for both networks, is on ETIS. Means over six seeds.

random flips. Each network is trained with and without FCT under six paired random seeds and tested on five public test sets, three of which are unseen in training. FCT improves ETIS for all three networks, by +1.65, +3.61 and +1.69 points, and CVC-ColonDB for PraNet-V2, by +3.01 points, all with paired 95% confidence intervals above zero. It raises the five-set mean Dice on every seed for Polyp-PVT and PraNet-V2, and on five of six for ASGNet, whose gain is borderline. FCT also lowers the five-set mean flip gap of Polyp-PVT and PraNet-V2 by 40% and 34%, respectively. On six more centres from the PolypGen benchmark [1], a label-aware variant improves the pooled Dice on every seed (FCT itself is pending). A component-by-component dissection relates these gains to the three supervised views, the pulls, their weight and the upright pull (four of its rows are still running).

Our contributions are summarized as follows:

- We show that, in binary segmentation under full supervision, a cross-entropy consistency loss between two views acts, pixel by pixel, as a reweighted supervised loss whose direction depends on which view is closer to the label.
- We derive from this identity FCT, a label-aware flip-consistency rule whose single constant is set by the decision threshold. It plugs into any binary segmentation network and adds no cost at inference.
- We evaluate FCT on three architectures, five test sets and six paired seeds with confidence intervals, and dissect it component by component. It improves ETIS for all three networks, raises the five-set mean Dice on every seed for two of them, and makes the predictions of both markedly more flip-consistent.

II. Related Work

A. Polyp Segmentation and Generalization

Deep polyp segmentation has advanced mainly through network design. PraNet [5] aggregates high-level features with a parallel partial decoder and refines boundaries with reverse attention. SANet [6] adds shallow attention and colour exchange, and UACANet [20] attends to the uncertain region of each prediction. Later networks adopt transformer encoders, as Polyp-PVT [7] and ASGNet [9] do, the latter guiding its features with spectral information, or add supervision, as PraNet-V2 [8] does with dual-supervised reverse attention. Other recent designs add contrastive transformer modules [21], boundary-sensitive attention [22] or uncertainty modeling [23], or adapt the Segment Anything Model with a convolutional branch [24]. Closest to our motivation, VANet [25] observes that steering the colonoscope tip changes the viewpoint and the appearance of polyps, and handles this with viewpoint-aware modules. These networks change the architecture, whereas FCT changes only the training loss, so it complements rather than replaces them.

Most of these models follow the protocol of [5]: they train on Kvasir-SEG [26] and CVC-ClinicDB [27], and they are tested on these two datasets and on CVC-ColonDB [28], CVC-300 [29] and ETIS [30], which are unseen in training. Accuracy on the unseen sets stays well below that on the seen ones, and multi-centre benchmarks confirm the gap at scale [1], [2]. Domain generalization offers training-side remedies [31], such as shape-aware meta-learning [32] and frequency-space interpolation across sites [33]. For polyps, randomized global illumination augmentation [34] and colour transfer [23] target nuisance factors that vary across centres, and domain adaptation learns from unlabeled images of the

target domain [35]. These methods diversify the training data, learn from differences between domains or need target images, whereas FCT adds a constraint between views of each training image and needs no target data, so the approaches are complementary.

B. Consistency Regularization and Equivariance

Consistency regularization asks a network to agree with itself across perturbed views of an input. It was developed for semi-supervised learning, where unlabeled images provide only this signal: the Π-model and temporal ensembling [10], mean teachers [11], unsupervised data augmentation [36] and FixMatch [12]. FixMatch keeps only pseudo-labels whose confidence exceeds a threshold, a proxy for correctness when the true label is unknown. Semi-supervised segmentation methods perturb inputs, features or networks [37]–[39]. In medical imaging, consistency under spatial transformations exploits the equivariance of segmentation on unlabeled scans [13], [14], or its invariance to the surrounding context [40]. Mean teachers restrict consistency to reliable regions [41] or to ambiguous ones [42], bidirectional patch displacement builds new samples for consistency learning guided by prediction confidence [43], and semi-supervised polyp segmentation aligns the affinity maps of a student and a teacher [44].

With labels available, consistency also serves as a regularizer: AugMix [15] matches predictions across augmentation chains, and R-Drop [16] across dropout masks. A sample-efficiency analysis shows that, with label-invariant augmentations, consistency regularization learns from fewer samples than training on the augmented copies with their labels [45]. Flip and rotation augmentation, by contrast, supervises each transformed copy with its own transformed label and does not relate the predictions on different copies. Equivariance can also be built into the architecture [46], [47], enforced on class activation maps in weakly supervised segmentation [48], or approximated at test time by averaging predictions over transformed inputs [49], at the price of a constrained architecture or a multiplied inference cost. All these methods treat the views as equally reliable or weigh them by the model's own confidence, since none uses the label to compare views; under full supervision, FCT does exactly that.

C. Self-Distillation and Label-Aware Weighting

Knowledge distillation trains a student to match the soft predictions of a teacher [50]. In self-distillation, the teacher is the model itself: an earlier generation of the same architecture [17], its own deeper layers [51], its predictions from the previous epoch [52] or dropout-sampled versions of itself [53]. Closest to our setting, data-distortion guided self-distillation [54] makes a network agree with itself on two distorted copies of an image under full supervision, treating both copies alike. The consistency term of FCT is also self-distillation across views, but it compares teacher and student against the label, pixel by pixel.

Several works relate distillation to the ground truth. Analyses of the distillation gradient show that it rescales the supervised gradient according to the teacher's confidence in the true class [17], [18], self-distillation over several rounds averages the labels of instances with correlated features [55], and Yuan et al. [19] relate distillation to label smoothing. Decoupled knowledge distillation [56] splits the loss exactly into target-class and non-target-class terms, knowledge adjustment [57], refined logits [58] and label revision [59] correct teacher predictions that contradict the label, and confidence-aware multi-teacher distillation [60] weights each teacher by how close its predictions are to the label. For segmentation, structured distillation transfers pixel-wise and pairwise knowledge from a large network to a compact one [61]. Loss reweighting also changes how much each example counts: focal loss [62] and online hard example mining [63] emphasize examples with high loss, and the structure loss used by PraNet-style networks emphasizes pixels near boundaries [64]. By our identity, FCT belongs to this family: it reweights the supervised gradient of each pixel, but by how the pixel's error compares with that of another view of the same pixel, within one network rather than across separate teachers.

III. When Does a Consistency Pull Help?

This section shows that, when the label is binary, a consistency loss toward a detached teacher view acts at every pixel as the supervised loss, rescaled by a factor that compares the errors of the two views. The sign of this factor tells, pixel by pixel, whether the pull helps or hurts, and Section IV builds FCT on it.

A. Notation

Let $x \in \mathbb{R}^{H\times W\times 3}$ denote an image, $y \in \{0,1\}^{H\times W}$ its polyp mask, and f a segmentation network that maps x to a logit map $z = f(x) \in \mathbb{R}^{H\times W}$; the predicted probabilities are $p = \sigma(z)$, where σ denotes the logistic function. Let T_h and T_v denote the horizontal and vertical flips. Each flip is its own inverse, so the prediction for a flipped view, brought back to the upright frame, is

$$z^k = T_k f(T_k x), \qquad k \in \{h, v\}, \tag{1}$$

and the upright view gives $z^o = f(x)$. A flip-equivariant network satisfies $z^h = z^v = z^o$. The flip gap of Section I measures departures from this equality on binary masks:

$$\Delta(x) = 1 - \frac{1}{2}\sum_{k\in\{h,v\}} D\big(\mathbf{1}[p^o > \tfrac{1}{2}],\, \mathbf{1}[p^k > \tfrac{1}{2}]\big), \tag{2}$$

where D denotes the Dice overlap of two masks and $\mathbf{1}[\cdot]$ the indicator function, applied pixel by pixel.

At a pixel i, the error of a view is $e_i = |p_i - y_i| \in [0,1]$, and the view is correct at the decision threshold of $\frac{1}{2}$ if $e_i < \frac{1}{2}$. We drop the index i when a statement holds for each pixel separately. A consistency loss pulls a student view s toward a teacher view t whose probability p_t is detached, that is, treated as a constant. We write

$$\ell(z_s; q) = -q \log \sigma(z_s) - (1-q)\log\big(1 - \sigma(z_s)\big) \tag{3}$$

for the binary cross-entropy of the student's logit z_s with a target $q \in [0,1]$. The consistency loss is $\ell(z_s; p_t)$ and the

supervised loss is $\ell(z_s; y)$, and their gradients with respect to the logit are

$$\frac{\partial \ell(z_s; p_t)}{\partial z_s} = p_s - p_t, \qquad \frac{\partial \ell(z_s; y)}{\partial z_s} = p_s - y. \tag{4}$$

B. Consistency as Reweighted Supervision

The two gradients in Eq. (4) are parallel whenever the label is binary: the consistency gradient is the supervised one, rescaled by how the errors of the two views compare.

Proposition 1: Let $y \in \{0, 1\}$, $p_s, p_t \in [0, 1]$ and $p_s \neq y$. Then

$$\frac{\partial \ell(z_s; p_t)}{\partial z_s} = \Big(1 - \frac{e_t}{e_s}\Big) \frac{\partial \ell(z_s; y)}{\partial z_s}, \tag{5}$$

where $e_s = |p_s - y|$ and $e_t = |p_t - y|$ denote the errors of the student and the teacher.

Proof: Because y is 0 or 1, every $p \in [0, 1]$ satisfies $p - y = (1 - 2y)|p - y|$: for $y = 0$, $p - y = p$, and for $y = 1$, $p - y = -(1 - p)$. Hence $p_s - p_t = (p_s - y) - (p_t - y) = (1 - 2y)(e_s - e_t) = (p_s - y)(1 - e_t/e_s)$, where the last step uses $(1 - 2y)e_s = p_s - y$ and $e_s > 0$. Eq. (4) completes the proof. ∎

The identity needs no assumption on the network or on the teacher's probability, only a binary label, which places both errors on the same side of the target. It also holds for a squared-error pull on probabilities, whose gradient with respect to p_s is again $p_s - p_t$. Summed over pixels and differentiated with respect to the network parameters θ, Eq. (5) reads

$$\nabla_\theta \sum_i \ell(z_{s,i}; p_{t,i}) = \sum_i \omega_i \, \nabla_\theta \, \ell(z_{s,i}; y_i), \qquad \omega_i = 1 - \frac{e_{t,i}}{e_{s,i}}, \tag{6}$$

with the weights ω_i held constant. A consistency loss is thus a supervised loss with per-pixel weights, and some of these weights are negative.

Corollary 1 (Lag condition): Under the conditions of Proposition 1, a gradient step on $\ell(z_s; p_t)$ moves the student's probability toward the label if $e_t < e_s$, away from it if $e_t > e_s$, and not at all if $e_t = e_s$.

Proof: A step of size η changes the logit by $-\eta(p_s - p_t) = -\eta \, \omega \, (p_s - y)$, by Eq. (5). The supervised step $-\eta(p_s - y)$ moves p_s toward y because σ is increasing, and $\omega > 0$ if and only if $e_t < e_s$. ∎

To first order, the step lowers the student's error by $\eta \, \sigma'(z_s)(e_s - e_t)$, so we call $e_s - e_t$ the pull toward the label. Fig. 2(a) maps it over the two errors. Below the diagonal, the teacher leads and the pull helps. Above it, the student leads and the pull hurts, in two ways. Where the teacher is wrong and the student right ($e_s < \frac{1}{2} < e_t$), the pull drives a correct pixel toward a misclassification, with a strength $e_t - e_s$ that can approach that of a full supervised step. Where both are right ($e_s < e_t < \frac{1}{2}$), the pull only lowers the student's confidence, and it stops when the student reaches the teacher, on the correct side of the threshold.

Corollary 1 answers the question of our title at the level of a pixel. In semi-supervised learning, no label separates the two cases, so the views are pulled toward each other alike. Under full supervision, the label separates them at every pixel, and a pull can be kept where the teacher leads, and elsewhere only where it cannot reverse a correct decision. Section IV turns this rule into a weight.

IV. Label-Aware Flip-Consistency Training

FCT turns Corollary 1 into a training objective. Each training image is presented in three views, every view is supervised with the label, and the views teach each other only where the label shows that the teacher is closer to it, or where a pull cannot reverse a correct decision. Fig. 3 gives an overview. This section describes the supervised views, the pulls on the flipped views, the pull on the upright view and the full objective in turn, and then relates FCT to other consistency rules.

A. Three Supervised Views

Each image is passed through the network upright and flipped horizontally and vertically, and the flipped outputs are flipped back to the upright frame as in Eq. (1). Every view is supervised with the host network's own loss:

$$\mathcal{L}_{\text{sup}} = \frac{1}{3} \sum_{k \in \{o,h,v\}} \mathcal{L}_{\text{host}}\big(Z^k, y\big), \tag{7}$$

where Z^k denotes all output maps of view k in the upright frame and $\mathcal{L}_{\text{host}}$ the supervised loss of the host network, unchanged; the prediction logit z^k is formed from Z^k as the host forms it at test time. The factor $\frac{1}{3}$ keeps the supervised loss at the scale of single-view training. On its own, Eq. (7) is a deterministic form of flip augmentation: each view is tied to the label, but no view is tied to another.

B. Lag-or-Margin Flip Pulls

Each flipped view is pulled toward the upright view, its teacher. Following Corollary 1, the pull keeps its full strength where the flipped view lags behind the upright view, and it is weakened where the flipped view leads:

$$w_i^k = \begin{cases} 1, & e_i^k \geq e_i^o, \\ \max\big(0,\, 1 - a\, e_i^o\big), & e_i^k < e_i^o, \end{cases} \tag{8}$$

where e_i^k and e_i^o denote the errors of the flipped and the upright view at pixel i, computed without gradient, and $a > 0$ is a constant. The pull on flip k averages the weighted consistency loss of Eq. (3) over the image:

$$\mathcal{C}_k = \frac{1}{HW} \sum_i w_i^k \, \ell\big(z_i^k; p_i^o\big), \tag{9}$$

with the upright probability p^o detached.

The constant a follows from the decision threshold. On a leading pixel, the pull moves the flipped view away from the label (Corollary 1), and Eq. (8) keeps it in proportion to the margin $1 - a\, e^o$. We ask two things of this margin. First, a pull toward a wrong teacher must vanish, $w = 0$ whenever $e^o > \frac{1}{2}$, which requires $a \geq 2$. Second, a pull toward a correct teacher must survive however small its margin, $w > 0$ whenever $e^o < \frac{1}{2}$, which requires $a \leq 2$. Hence $a = 2$, and on a leading pixel with a correct teacher, $w = 1 - 2e^o = |2p^o - 1|$ is twice

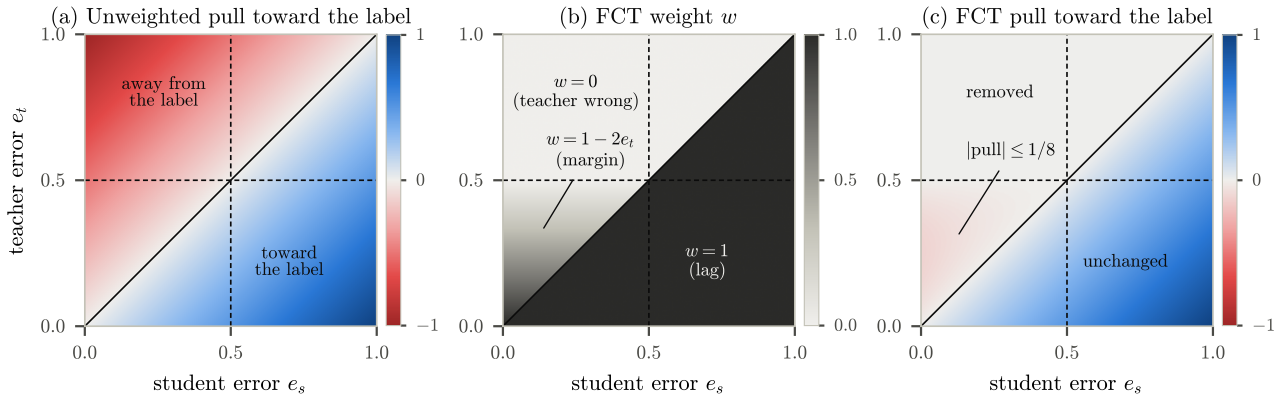


Fig. 2. Pull toward the label exerted on a student view by a detached teacher view at one pixel, as a function of the student's error e_s and the teacher's error e_t (Proposition 1); dashed lines mark the decision threshold. (a) An unweighted pull, $e_s - e_t$, pushes the student away from the label wherever it leads, even toward a wrong teacher (top left). (b) The FCT weight of Eq. (8). (c) With it, every pull toward the label is kept, every pull that moves the student away from the label toward a wrong teacher is removed, and the remaining pulls away from the label are bounded by $1/8$.

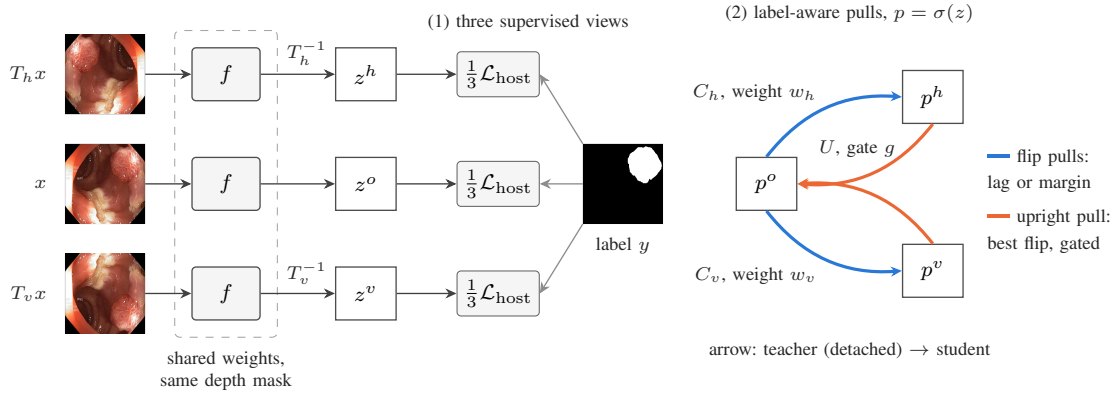


Fig. 3. Overview of the proposed FCT. (1) An image and its horizontal and vertical flips pass through one network; the flipped outputs are flipped back, and each view is supervised with the host network's own loss, weighted $\frac{1}{3}$. (2) The views then teach each other, with the teacher's probability detached: the upright view pulls each flipped view with the lag-or-margin weight of Eq. (8), and the flipped view closest to the label pulls the upright view wherever it is more accurate. At test time, only the upright view is used.

the teacher's distance from the threshold, that is, its margin rescaled to $[0, 1]$. The constant is thus set by the threshold rather than tuned, and Section V-G shows that the results are insensitive to it.

With $a = 2$, the pull on a leading pixel is weak. Its strength $w\,(e^o - e^k)$ is at most $(1 - 2e^o)\,e^o \le \frac{1}{8}$, against up to one for a supervised step (Fig. 2(c)), and it vanishes when the flipped view reaches its teacher, on the correct side of the threshold. It thus keeps the two views consistent where both are right, as flip equivariance asks, without reversing a decision by itself.

C. Upright Pull

Eq. (8) lets the upright view teach but never learn, although it is the view that the network sees at test time. The upright view is therefore also pulled toward the flipped view closest to the label, wherever that view leads:

$$\mathcal{U} = \frac{1}{HW} \sum_i g_i\, \ell\big(z_i^o; p_i^*\big), \qquad g_i = \mathbf{1}\big[e_i^* < e_i^o\big], \tag{10}$$

where p_i^* denotes the detached probability of the flipped view with the smaller error at pixel i, and e_i^* its error. By Corollary 1, every pull in Eq. (10) moves the upright view toward the label, so it needs no margin term.

D. Objective and Training

The full objective adds the pulls to the supervised loss:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \frac{\lambda}{2}\big(\mathcal{C}_h + \mathcal{C}_v\big) + \frac{\lambda}{2}\mathcal{U}, \tag{11}$$

where $\lambda \ge 0$ weighs consistency against supervision. The two flip pulls share λ, and the upright pull receives the weight of one flip pull. The weight is ramped up linearly at the start of training,

$$\lambda_t = \lambda \,\min\big(1,\, t/t_0\big), \tag{12}$$

where t counts the training batches and t_0 is the length of the ramp; we use $\lambda = 0.5$ and $t_0 = 300$.

Three implementation choices keep the views comparable and the memory low. First, the three views of an image share

one stochastic-depth mask [65], so that their differences reflect the flip rather than the random depth. Second, when a host trains at several input scales, FCT is applied at every scale. Third, the probabilities p^h and p^v of Eq. (10) come from passes without gradient that leave the running statistics of batch normalization unchanged. In our current implementation, these passes run before the upright pass and the graph of each view is released after its backward pass, so only one view's computation graph is held in memory at a time; the runs reported in Section V computed them after the upright pass, which gives the same objective at a higher peak memory. For each image and input scale, FCT thus runs five forward passes, three of them with gradients. At test time, the network is applied to the upright image only, exactly as without FCT, so FCT changes neither the architecture nor its inference cost.

E. Relation to Other Consistency Rules

Eq. (8) contains the usual rules as special cases or near neighbours, which Section V-G compares. With $w = 1$ everywhere, $\mathcal{C}_k$ is the unweighted transformation consistency of semi-supervised segmentation [13], [14], applied here to labeled images; by Eq. (6), it mixes pulls toward and away from the label. Confidence thresholding [12] keeps pulls whose teacher is confident, which includes confidently wrong teachers, the case that Corollary 1 rules out. Finally, a rule can use the label to judge the teacher alone, for example by weighting each pixel by whether and how confidently the upright view is correct. Such a rule ignores the student's error, so it cannot tell the lagging pixels, where a pull helps, from the leading ones, where it hurts. Of these rules, only FCT compares the two errors, as Corollary 1 requires.

V. Experiments

We test the claims of Sections III and IV in the order in which they were made: FCT improves all three networks over their own baselines, most on unseen data (Section V-C), and a label-aware variant generalizes to six more centres (Section V-D); FCT makes predictions more flip-consistent (Section V-E); a per-image analysis shows where its gains come from (Section V-F); and a dissection of its components shows that its weight acts on the gradient as Corollary 1 predicts and that its constant needs no tuning (Section V-G). Section V-H reports its cost.

A. Datasets and Evaluation Protocol

We train on 1,288 images, 800 from Kvasir-SEG [26] and 488 from CVC-ClinicDB [27], drawn from the 1,450-image training split of [5]; none of them appears in a test set. Because this set is smaller than the full split, absolute scores are not directly comparable with published ones, and every comparison below is between models trained on the same images. We test on the five standard test sets: the test splits of Kvasir-SEG (100 images) and CVC-ClinicDB (62), which we call seen, and CVC-ColonDB [28] (380), CVC-300 [29] (60) and ETIS [30] (196), which are unseen in training. CVC-ColonDB and CVC-300 come from the same group as CVC-ClinicDB, whereas ETIS comes from another centre.

We report the mean Dice (mDice) of the evaluation toolbox of [5], which Polyp-PVT also uses: each predicted map is min-max normalized and binarized at 256 thresholds, and the Dice coefficient is averaged over images and thresholds. The flip gap of Eq. (2) is computed on the masks thresholded at 0.5 at the input resolution of the network, with add-one smoothing so that two empty masks agree. Each network is evaluated on the five test sets after every epoch with the per-image Dice of the official Polyp-PVT code, and we keep the epoch with the highest five-set mean, for vanilla and FCT alike; Section V-C also reports the gains without this selection.

Each configuration is trained with six random seeds, and vanilla and FCT share each seed. We report the mean and standard deviation (SD) over seeds, the mean paired difference with its 95% confidence interval (CI) from the t-distribution with five degrees of freedom, and the number of seeds on which FCT scores higher. An exact permutation test over all 924 ways of splitting the twelve runs of a comparison into two groups of six reaches the same conclusion as the CI for every entry of Tables I and II.

B. Implementation Details

We add FCT to three networks without changing their architectures or retuning their recipes. Polyp-PVT [7] uses a PVTv2-B2 encoder [66]. As in its official code, it is trained with AdamW, a learning rate of 10^{-4}, a weight decay of 10^{-4}, batches of 16 images of 352×352 pixels, input scales of 0.75, 1 and 1.25, element-wise gradient clipping at 0.5 and 99 epochs, and its encoder has a stochastic-depth rate of 0.1; we add mixed precision. Its official recipe applies no orientation augmentation by default, and we keep this default, so vanilla Polyp-PVT never sees a flipped image. PraNet-V2 [8] uses a Res2Net-50 encoder [67], which has no stochastic depth, and is trained with Adam, a constant learning rate of 10^{-4}, batches of 8 images and 29 epochs, with the data pipeline, input scales and mixed precision of Polyp-PVT. ASGNet [9] uses a PVTv2-B3 encoder and keeps its own recipe: Adam, batches of 40 images accumulated over eight parts, 79 epochs at a single scale in full precision, a cosine learning-rate schedule, and its own data loader, which applies random horizontal flips, crops, rotations, colour changes and label noise. Unlike the other two, ASGNet is therefore trained with flip augmentation even without FCT. FCT uses $\lambda = 0.5$, set in preliminary runs on Polyp-PVT, and $a = 2$ for all three networks. Each run uses one [GPU model] GPU.

C. Main Results

FCT improves the five-set mean Dice of Polyp-PVT and PraNet-V2 on every seed and that of ASGNet on five of six, and for every network its largest gain is on an unseen set (Tables I–III, Fig. 4). For Polyp-PVT, the five-set mean rises by 0.52 points, with a 95% CI of $[0.15, 0.90]$, and ETIS gains 1.65 points $[0.88, 2.42]$ on all six seeds; the gains on the other four sets are positive on average but within seed variation. For PraNet-V2, the five-set mean rises by 1.68 points $[0.69, 2.67]$. Its largest gains are on ETIS, 3.61 points $[0.23, 6.99]$, and on CVC-ColonDB, 3.01 points $[1.25, 4.77]$ on all six seeds, and

TABLE I
POLYP-PVT (PVTV2-B2) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	91.94 ± 0.59	92.50 ± 0.41	+0.56	[−0.31, 1.44]	4/6
CVC-ClinicDB	93.17 ± 0.64	93.36 ± 0.23	+0.19	[−0.64, 1.01]	4/6
Unseen					
CVC-ColonDB	80.28 ± 0.68	80.43 ± 0.60	+0.15	[−0.80, 1.10]	3/6
CVC-300	88.79 ± 0.86	88.85 ± 1.10	+0.06	[−1.39, 1.51]	2/6
ETIS	78.61 ± 0.29	80.26 ± 0.65	**+1.65**	**[0.88, 2.42]**	6/6
Mean	86.56 ± 0.17	87.08 ± 0.32	**+0.52**	**[0.15, 0.90]**	6/6

Training set: 1,288 Kvasir and CVC-ClinicDB images; *seen*: test splits of the two training datasets; *unseen*: datasets not used in training. Gain: mean paired difference, FCT minus vanilla with the same seed. 95% CI: t-distribution, five degrees of freedom. Seeds: seeds, of six, on which FCT scores higher. Mean: average over the five test sets.

TABLE II
PRANET-V2 (RES2NET-50) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	87.94 ± 1.46	89.63 ± 0.55	**+1.69**	**[0.12, 3.26]**	5/6
CVC-ClinicDB	89.80 ± 1.30	89.83 ± 0.86	+0.03	[−1.31, 1.38]	3/6
Unseen					
CVC-ColonDB	69.91 ± 2.30	72.92 ± 1.27	**+3.01**	**[1.25, 4.77]**	6/6
CVC-300	87.73 ± 2.92	87.76 ± 1.47	+0.04	[−3.31, 3.39]	3/6
ETIS	67.07 ± 1.72	70.68 ± 1.69	**+3.61**	**[0.23, 6.99]**	5/6
Mean	80.49 ± 0.93	82.17 ± 0.64	**+1.68**	**[0.69, 2.67]**	6/6

Columns and statistics as in Table I.

TABLE III
ASGNET (PVTV2-B3) WITH AND WITHOUT FCT ON FIVE POLYP TEST SETS: MDICE (%), MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Test set	Vanilla	FCT (ours)	Gain	95% CI	Seeds
Seen					
Kvasir	90.86 ± 0.15	90.80 ± 0.33	−0.06	[−0.46, 0.34]	1/6
CVC-ClinicDB	91.09 ± 1.11	90.87 ± 0.75	−0.21	[−0.96, 0.53]	2/6
Unseen					
CVC-ColonDB	80.41 ± 0.71	80.98 ± 1.04	+0.57	[−0.72, 1.85]	4/6
CVC-300	89.40 ± 0.41	88.96 ± 0.97	−0.44	[−1.50, 0.62]	3/6
ETIS	77.44 ± 1.23	79.12 ± 0.85	**+1.69**	**[0.08, 3.30]**	5/6
Mean	85.84 ± 0.42	86.15 ± 0.33	+0.31	[0.00, 0.61]	5/6

Columns and statistics as in Table I. ASGNet's own recipe already trains with random horizontal flips. Its five-set mean gain is borderline: the CI excludes zero by less than 0.01 points (paired t-test, $p = 0.049$), and the exact permutation test gives $p = 0.20$.

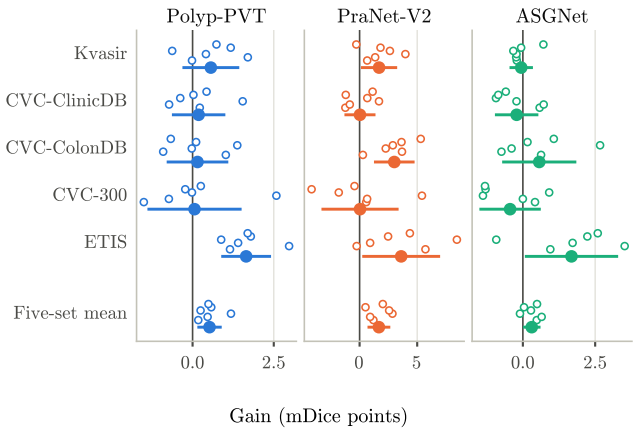


Fig. 4. Paired gains of FCT over vanilla in mDice points: one hollow marker per seed, and the mean with its 95% CI below them. Every seed gains on the five-set mean for Polyp-PVT and PraNet-V2 and five of six for ASGNet; the largest gains are on ETIS and, for PraNet-V2, CVC-ColonDB.

Kvasir also improves, by 1.69 points $[0.12, 3.26]$. For PraNet-V2, FCT also lowers the SD over seeds on every set. For these two networks, no set loses on average; the smallest gains are on CVC-300 for both, $+0.06$ and $+0.04$ points, and on CVC-ClinicDB for PraNet-V2, $+0.03$. ASGNet, whose own recipe already trains with random horizontal flips, gains least (Table III). Its ETIS score rises by 1.69 points $[0.08, 3.30]$ on five of six seeds, but its five-set mean rises by only 0.31 points: the CI excludes zero by less than 0.01 points (paired $p = 0.049$), and the exact permutation test does not support the gain ($p = 0.20$). Its other four sets change by less than 0.6 points in either direction, within seed variation.

Without epoch selection, the gains remain positive on average but are less certain. Fig. 5 shows the five-set mean of the training-time evaluator after every epoch: the six-seed mean of FCT lies above that of vanilla after 81 of 99 epochs for Polyp-PVT, 28 of 29 for PraNet-V2 and 76 of 79 for ASGNet. Averaged over the last ten epochs, without any selection, FCT gains $+0.42$ $[-0.03, 0.88]$, $+2.41$ $[1.40, 3.43]$ and $+0.21$ $[-0.59, 1.01]$ points on the five-set mean of the three networks, and $+1.52$ $[0.11, 2.93]$, $+3.71$ $[0.75, 6.68]$ and $+1.99$ $[-0.26, 4.24]$ points on ETIS (Appendix, Table XI). At the last epoch alone, no five-set mean gain has a CI above zero: $+0.55$ $[-0.11, 1.22]$ for Polyp-PVT, $+0.49$ $[-1.17, 2.14]$ for PraNet-V2, whose curves peak early and decline, so that its last epoch is a poor checkpoint for either arm, and $+0.13$ $[-0.91, 1.17]$ for ASGNet.

D. Six More Centres

Of the five standard test sets, only ETIS comes from another centre than the training images (Section V-A). PolypGen [1] adds six more centres; we use its 1,411 single-frame images with a polyp, none of which took part in training or in selecting a checkpoint. We fixed the analysis before scoring any map: the primary result is the mDice pooled over the six centres, every image counted once, over all six seeds, and each centre is a secondary result. So far, PolypGen maps exist only for the confidence-graded weight, row (h) of Table VII, and not yet for FCT. This label-aware variant raises the pooled mDice of Polyp-PVT from 82.14% to 82.98%, by 0.84 points $[0.59, 1.09]$, on all six seeds, and its gain has a CI above zero on four of the six centres (Table IV); an exact permutation test over the 924 splits of the twelve runs gives $p = 0.002$ for the pooled gain. FCT's checkpoints will be scored with the same analysis.

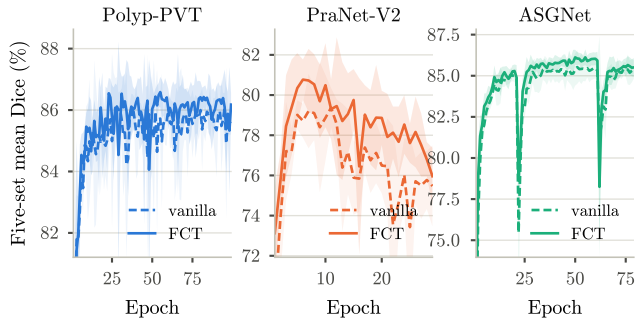


Fig. 5. Five-set mean Dice of the training-time evaluator after every epoch, without (dashed) and with (solid) FCT; mean and SD over six seeds. FCT lies above vanilla through most of training. Every ASGNet run, with or without FCT, drops briefly at epoch 62, so the drop comes from its recipe, not from FCT.

TABLE IV
SIX MORE CENTRES (POLYPGEN): MDICE (%) OF VANILLA POLYP-PVT AND OF ITS CONFIDENCE-GRADED VARIANT, ROW (H) OF TABLE VII, MEAN ± SD OVER SIX SEEDS. BOLD: GAINS WHOSE 95% CI EXCLUDES ZERO.

Centre	Images	Vanilla	Variant (h)	Gain [95% CI]	Seeds
C1	251	86.65 ± 0.42	87.94 ± 0.55	**+1.29 [0.71, 1.86]**	6/6
C2	270	84.60 ± 0.47	84.31 ± 0.64	−0.28 [−1.27, 0.70]	2/6
C3	455	89.93 ± 0.24	90.46 ± 0.20	**+0.53 [0.25, 0.81]**	6/6
C4	146	68.33 ± 0.99	69.23 ± 1.47	+0.90 [−1.63, 3.44]	4/6
C5	206	65.71 ± 1.65	67.92 ± 1.25	**+2.21 [0.38, 4.04]**	5/6
C6	83	82.88 ± 0.29	84.23 ± 0.45	**+1.35 [1.04, 1.65]**	6/6
Pooled	1411	82.14 ± 0.18	82.98 ± 0.21	**+0.84 [0.59, 1.09]**	6/6

Single-frame images with a polyp; none was used in training or in selecting a checkpoint. Pooled: every image of the six centres counted once, the primary result of an analysis fixed before scoring. FCT itself: pending.

E. Flip Consistency

FCT makes both networks more flip-consistent on every test set (Table V). It lowers the five-set mean flip gap from 12.5% to 7.5% for Polyp-PVT and from 15.6% to 10.2% for PraNet-V2, and the flip gap on the training images by 84% and 76%. The reduction holds on every seed and in every row of the table. Two observations qualify it. First, flip consistency is not accuracy: the largest reduction on a test set, 70% for Polyp-PVT on CVC-ClinicDB, comes with almost no change in Dice, and CVC-ColonDB and ETIS remain the least consistent sets after FCT. Second, training on flipped images lowers the flip gap by itself. In preliminary single runs of Polyp-PVT, supervising the three views without any pull, row (b) of Table VII, reached a five-set flip gap of 8.2%, and the official random flip and rotation augmentation reached 6.8%, below that of FCT; yet the five-set mean Dice of the latter, 85.75%, was below that of every vanilla seed. A lower flip gap alone therefore does not explain the gains of FCT, and we read it as a symptom that the views are tied together, not as the cause of the gains.

F. Where Do the Gains Come From?

For PraNet-V2, the gains on the unseen sets are spread over many images: on ETIS, the 176 images outside the ten largest gains and the ten largest losses contribute 2.94 of its 3.61 points. For Polyp-PVT, they are more concentrated: on ETIS, the ten images with the largest gains contribute 1.78 points, more than the net gain of 1.65, and the other 176 images contribute 0.79. Table VI breaks the errors down at a threshold of 0.5 on the per-image min-max-normalized maps. The two networks gain in different ways. PraNet-V2 fails less often: the number of images with a Dice below 0.1 falls from 31.8 to 25.0 per run on ETIS and from 56.5 to 47.5 on CVC-ColonDB. Polyp-PVT fails about as often with FCT as without, 10.3 and 10.5 images per run on ETIS; it gains instead by predicting less false-positive area, 1.31 instead of 1.52 times the polyp area. In two cases, small polyps carry much of the gain: the quarter of images with the smallest polyps gains 5.46 points on ETIS for Polyp-PVT and 6.16 points on CVC-ColonDB for PraNet-V2. The pattern is not general: on ETIS, the smallest polyps gain 2.70 points for PraNet-V2 and the next quarter 6.05, and on CVC-300 the smallest polyps lose with both networks. These breakdowns are descriptive, and with four size groups per set and network, some differences are expected by chance.

Fig. 6 shows examples. Without FCT, PraNet-V2 misses the polyp in rows 1, 2 and 4, and Polyp-PVT adds spurious

TABLE V
FLIP GAP (%) OF THE PREDICTED MASKS WITHOUT AND WITH FCT ON EACH TEST SET AND ON THE TRAINING IMAGES, AND ITS RELATIVE CHANGE. MEANS OVER SIX SEEDS; LOWER IS BETTER.

	Polyp-PVT			PraNet-V2		
Images	Vanilla	FCT	Change	Vanilla	FCT	Change
Seen						
Kvasir	6.3	4.1	−35%	8.8	5.1	−42%
CVC-ClinicDB	12.4	3.7	−70%	13.1	5.9	−55%
Unseen						
CVC-ColonDB	16.9	11.3	−33%	20.4	14.8	−27%
CVC-300	9.1	5.1	−43%	9.8	6.2	−37%
ETIS	17.8	13.2	−26%	25.8	19.2	−26%
Five-set mean	12.5	7.5	−40%	15.6	10.2	−34%
Training images	6.3	1.0	−84%	8.5	2.0	−76%

Flip gap: one minus the Dice overlap of the binary masks predicted for an image and for its flipped copy, flipped back, averaged over horizontal and vertical flips and over images. FCT lowers it on every seed, in every row.

TABLE VI
WHERE THE GAINS COME FROM ON THE UNSEEN TEST SETS, VANILLA → FCT: FAILURES, MISSED AND FALSE-POSITIVE AREA, AND THE GAIN ON THE SMALLEST POLYPS. MEANS OVER SIX SEEDS.

Test set	Gain	Failures	Missed area	False-pos. area	Smallest
Polyp-PVT					
CVC-ColonDB	+0.15	15.0 → 15.3	0.18 → 0.16	0.66 → 0.73	−1.75
CVC-300	+0.06	0.2 → 0.0	0.06 → 0.06	0.23 → 0.27	−2.38
ETIS	+1.65	10.5 → 10.3	0.12 → 0.11	1.52 → 1.31	+5.46
PraNet-V2					
CVC-ColonDB	+3.01	56.5 → 47.5	0.27 → 0.25	0.33 → 0.26	+6.16
CVC-300	+0.04	0.2 → 0.0	0.05 → 0.06	0.25 → 0.23	−1.85
ETIS	+3.61	31.8 → 25.0	0.26 → 0.22	0.82 → 0.86	+2.70

Gain: mDice points. Failures: images per run with Dice below 0.1 for the mask thresholded at 0.5 after per-image min-max normalization. Missed and false-positive area: at the same threshold, as fractions of the polyp area. Smallest: mDice gain on the quarter of the set with the smallest polyps.

regions in rows 1–3; with FCT, both predictions concentrate on the polyp. The last row shows the image on which FCT loses most: with FCT, Polyp-PVT adds a false region, and PraNet-V2 misses the polyp.

G. Ablation Studies

Component dissection. FCT has four components: the three supervised views of Eq. (7), the pulls on the flipped views, their label-aware weight of Eq. (8), and the upright pull of Eq. (10). Table VII dissects them on Polyp-PVT, against the same vanilla runs throughout. Row (b) supervises the three views without any pull. It is the flip-augmentation control, and row (b) minus vanilla is the gain of the flips alone. Rows (c)–(e) add to row (b) the upright pull alone, the weighted flip pulls alone, and both, which is FCT. Their differences from row (b) decompose the gain of FCT over flip augmentation, and comparing row (e) with the two single additions shows whether the pulls add up. Rows (f) and (g) change only the weight function of FCT, to unweighted flip pulls and to the lag gate without its margin. We fixed the protocol before seeing these rows: rows (b) and (c) are trained with six seeds, and rows (f) and (g) with three seeds first, extended to six only if they land near FCT. Rows (b), (c), (f) and (g) are being trained. ASGNet already offers a second test of FCT beyond flip augmentation: its own recipe trains with random horizontal flips, and FCT still raises its ETIS score by 1.69 points (Table III), although its five-set mean gain is borderline. Preliminary single unseeded runs of row (b) and of unweighted flip pulls without the upright pull reached five-set means of 86.56% and 87.06% and ETIS scores of 80.78% and 80.49%, against 87.08% and 80.26% for FCT over six seeds; single runs cannot rank variants that differ by less than a point. With six seeds, the paired CI of the difference between two variants spans about ± 0.4 to ± 0.7 points on the five-set mean, and with three seeds about 2.4 times more, so small effects will not be resolved.

The completed rows already answer two questions. First, the upright pull adds little on top of the weighted flip pulls: FCT minus row (d) is $+0.05$ $[-0.36, 0.47]$ points on the five-set mean, although FCT is the only variant so far that improves all five sets on average. Second, the form of the label-aware weight barely moves Dice. Row (h) replaces the lag-or-margin weight with the confidence-graded weight sketched in Section IV-E, which keeps a weight of 1 where the upright view is correct, raises it to $1 + p^o$ on missed polyp pixels and lowers it to $1 - p^o$ on false-positive pixels; row (d) minus row (h) is $+0.06$ $[-0.60, 0.73]$. Rows (d), (e) and (h) all improve ETIS with CIs above zero.

Their gradients, however, differ sharply. The gradient of the lag-or-margin pulls has a cosine of 0.53 with the gradient of the rest of the objective, against 0.26 for the confidence-graded weight, and it is about a third smaller relative to it: the ratio of the two gradients' norms is 0.08, against 0.12. The gap holds throughout training (Fig. 7) and on every seed: the cosine ranges over $[0.47, 0.56]$ for rows (d) and (e) and over $[0.18, 0.33]$ for row (h). This is the effect that Eq. (6) predicts. The confidence-graded weight keeps full weight on the pulls of leading flipped views, which point away from the label (Corollary 1), and even raises the weight where the upright view misses a polyp. Leading pixels are common: in the runs where we logged them, Polyp-PVT with $a < 2$ and PraNet-V2, a flipped view leads the upright view on about 44% of its pixels at the end of training, although this is a share of pixels, not of the gradient. Row (f) will show whether the weight's effect on the gradient carries over to Dice.

Constant a. The results are insensitive to a, so the constant needs no tuning (Table VIII; row (g) of Table VII, pending, adds the limit $a \to \infty$, which keeps only the lagging pulls). Every value in $\{0.5, 1, 1.5, 2\}$ improves ETIS by 1.6 to 2.1 points, and none differs significantly from $a = 2$ on the five-set mean. The value $a = 2$ is the only one that improves all five sets on average, but this distinction is within seed variation. These runs do not fully explain the insensitivity. The case that $a = 2$ removes, a flipped view that leads a wrong upright view, is rare: it arises on only 0.16% of the pixels at the end of training (Polyp-PVT, $a < 2$) and on 0.24% (PraNet-V2). But a also scales the pull on the far more frequent leading pixels with a correct upright view, and this change, too, leaves Dice unchanged within seed variation.

H. Cost

FCT multiplies the training time per epoch by 3.6 for Polyp-PVT, 3.3 for PraNet-V2 and 3.7 for ASGNet (Table IX). This matches its five forward and three backward passes for each image and input scale, which cost $11/3 \approx 3.7$ times a single view if a backward pass costs twice a forward pass. Without the upright pull, the two passes without gradient are not needed, and the measured factor drops to 3.0. Inference is unchanged: the trained network is the host network, applied once to the upright image.

VI. Discussion and Limitations

Why the unseen sets? For all three networks, FCT gains most on ETIS, the test set from another centre, and, for PraNet-V2, it also gains on CVC-ColonDB. These are the two sets on which every network is least accurate, and on which Polyp-PVT and PraNet-V2 are least flip-consistent. One explanation is the hypothesis of Section I: tying the views together discourages cues that change under a flip, and such cues are unlikely to transfer to a new centre. Our data support this only indirectly. The flip gap falls on every set, but its fall does not predict where Dice rises, and a single run with flip and rotation augmentation lowered it without any gain. The experiments do not yet separate three candidate causes of the gains: training on the supervised flipped views, flip consistency itself, and a consistency gradient better aligned with supervision. Preliminary single runs (Section V-G) leave open that the first two account for much of the gain; the pending rows of the component dissection (Table VII) are designed to settle this.

Scope of the analysis. Proposition 1 needs a binary label and a cross-entropy or squared-error pull. It applies to each class of a multi-label task, one class against the rest, but not directly to a softmax over several classes, where the two error

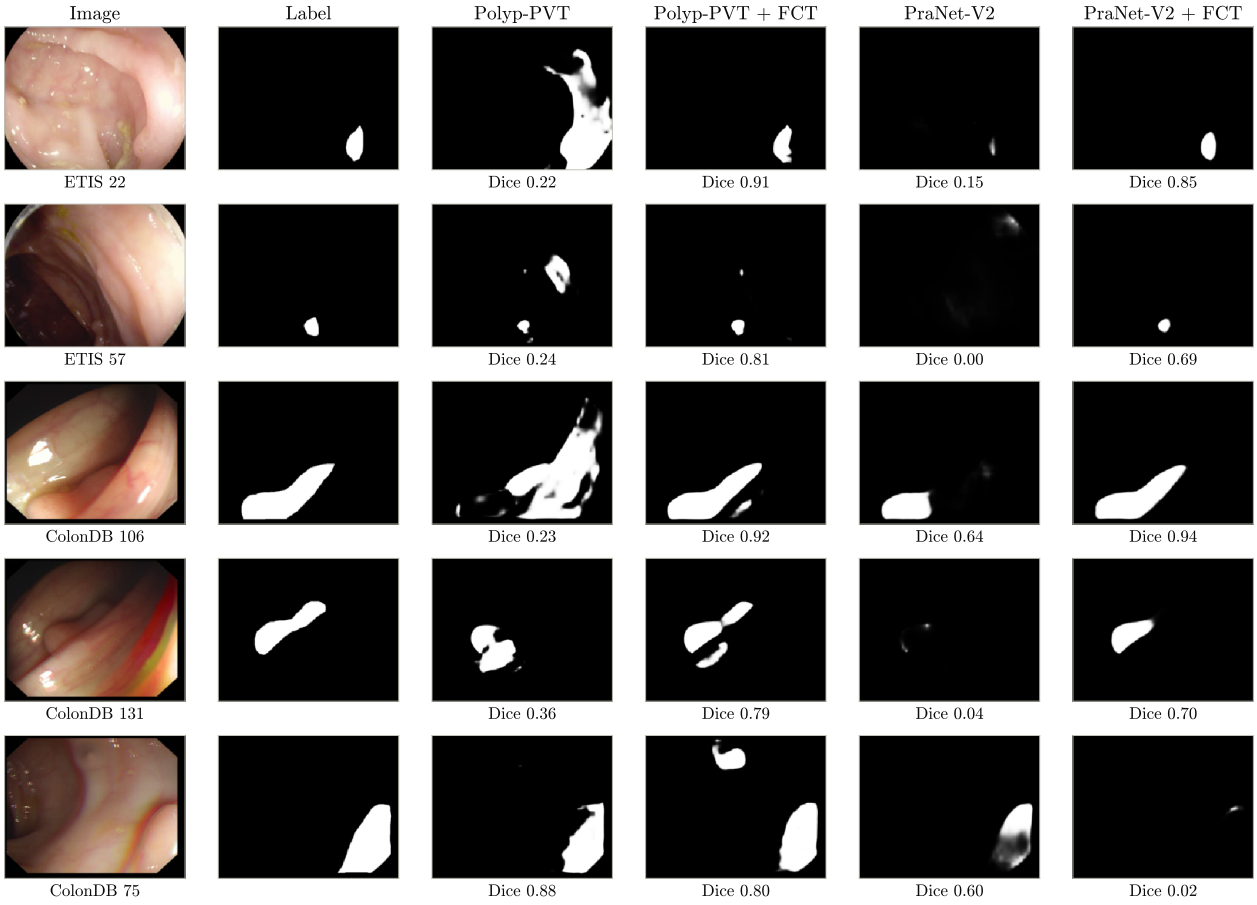


Fig. 6. Examples from the unseen sets: image, label, and the maps predicted by each network without and with FCT (seed 1), each with its mDice. Rows 1–4: four of the twelve unseen-set images with the largest gains, averaged over six seeds and both networks. Row 5: the image with the largest loss.

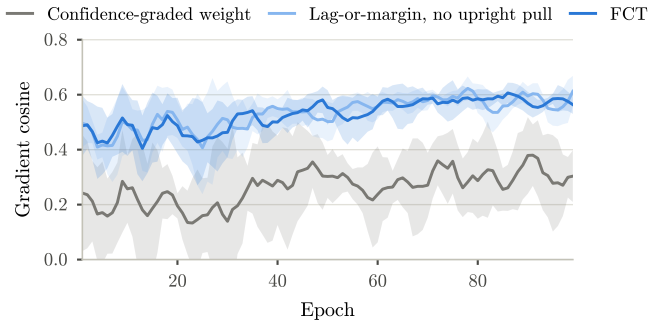


Fig. 7. Cosine between the gradient of the flip pulls and that of the rest of the objective during training of Polyp-PVT, for rows (d), (e) and (h) of Table VII; per-epoch means, smoothed over five epochs, with the SD over six seeds. The lag-or-margin weight doubles the alignment of the consistency gradient with the rest of the objective.

vectors need not be parallel. Soft labels also break it, since the two errors may then lie on opposite sides of the target; FCT therefore binarizes the label at 0.5 when it computes the weights, including at rescaled input sizes, where interpolation softens the mask boundary. The analysis also stops at the logits: it does not describe how the network shares the per-pixel pulls through its parameters.

Cost. FCT multiplies the training time by about 3.5 and leaves inference unchanged. Without the upright pull, the factor drops to three (Table IX) at a similar accuracy (Table VII); applying FCT to part of each batch would lower it further.

Limits of the evidence. The evidence has five limits. First, it covers three networks and one training set of 1,288 images; of the five standard test sets, only ETIS comes from another centre, and on the six PolypGen centres only the confidence-graded variant has been tested so far. Second, the reported epoch is selected on the test sets. The selection is the same for both arms, and the gains remain positive on average without it but are not significant at the last epoch (Fig. 5); the absolute scores are optimistic. Third, the weight was developed over several preliminary variants scored on the same five test sets, so these sets also guided its design. The confidence-graded weight, for instance, was carried to six seeds because of its result on seed 1, which row (h) of Table VII includes; on seeds 2–6 alone, its five-set mean gain is $+0.32$ $[0.08, 0.56]$. The gain of FCT is smallest on seed 1, $+0.18$ points, and is $+0.59$ $[0.15, 1.03]$ on seeds 2–6. Fourth, six seeds resolve gains of about half a point on the five-set mean, but not the smaller effects on single sets, whose CIs are up to several points wide. Fifth, λ was set on Polyp-PVT and not retuned for

TABLE VII
COMPONENT DISSECTION OF FCT ON POLYP-PVT: PAIRED MDICE GAINS (POINTS, 95% CI), THE EFFECT OF EACH COMPONENT, TEST SETS IMPROVED, FLIP GAP AND GRADIENT ALIGNMENT. BOLD: CIS THAT EXCLUDE ZERO.

Variant	Views	Flip pulls	Upright pull	Mean gain	ETIS gain	Component effect	Sets	Flip gap (%)	Cosine
Building FCT on the three supervised views									
(a) Vanilla	–	–	–	–	–	–	–	12.5	–
(b) Three views	✓	–	–	pending	pending	–	–	–	–
(c) + upright pull	✓	–	✓	pending	pending	pending	–	–	–
(d) + flip pulls	✓	Eq. (8)	–	+0.47 [−0.02, 0.96]	**+1.97 [1.06, 2.87]**	pending	4/5	7.5	0.53
(e) + both (FCT)	✓	Eq. (8)	✓	**+0.52 [0.15, 0.90]**	**+1.65 [0.88, 2.42]**	pending	5/5	7.5	0.53
Changing the weight function of FCT									
(f) FCT, $w = 1$ (3 seeds)	✓	$w = 1$	✓	pending	pending	pending	–	–	–
(g) FCT, lag only (3 seeds)	✓	lag only	✓	pending	pending	pending	–	–	–
Another label-aware weight, without the upright pull									
(h) Graded weight	✓	graded	–	**+0.40 [0.13, 0.68]**	**+1.56 [0.53, 2.60]**	+0.06 [−0.60, 0.73]	3/5	7.7	0.26

Six seeds unless the row says otherwise. Mean and ETIS gain: over vanilla with the same seeds. Component effect, on the five-set mean, paired by seed: for rows (c)–(e), the row minus row (b), so the effect of the pulls added to the three views; for rows (f) and (g), FCT minus the row, the effect of FCT's weight function; for row (h), row (d) minus row (h). Sets: test sets, of five, with a positive mean gain. Cosine: between the gradient of the flip pulls and that of the rest of the objective, measured every 50 batches and averaged over training and seeds; for row (e), the rest includes the upright pull. Lag only: $w = 1$ where the flipped view lags and 0 where it leads ($a \to \infty$). Graded: the confidence-graded weight of Section V-G.

TABLE VIII
SENSITIVITY TO THE CONSTANT a OF EQ. (8) ON POLYP-PVT: PAIRED MDICE GAINS OVER VANILLA AND OVER $a = 2$ (POINTS, SIX SEEDS, 95% CI). BOLD: CIS THAT EXCLUDE ZERO.

a	Mean gain	ETIS gain	Sets	$a = 2$ minus row
0.5	**+0.51 [0.30, 0.73]**	**+2.15 [1.37, 2.92]**	3/5	+0.01 [−0.47, 0.49]
1	+0.33 [−0.02, 0.67]	**+1.58 [0.40, 2.77]**	3/5	+0.19 [−0.26, 0.65]
1.5	**+0.36 [0.02, 0.71]**	**+1.83 [0.83, 2.82]**	3/5	+0.16 [−0.24, 0.56]
2 (FCT)	**+0.52 [0.15, 0.90]**	**+1.65 [0.88, 2.42]**	5/5	–

Gains over vanilla on the five-set mean and on ETIS. Sets: test sets, of five, with a positive mean gain. Last column: paired difference on the five-set mean between $a = 2$ and the row.

TABLE IX
COST OF FCT: TRAINING TIME PER EPOCH (MINUTES, MEDIAN OVER SIX SEEDS) AND INFERENCE COST.

Network	Vanilla	FCT	Ratio	Inference
Polyp-PVT	0.56	2.01	3.6×	unchanged
PraNet-V2	0.93	3.11	3.3×	unchanged
ASGNet	1.01	3.70	3.7×	unchanged
Polyp-PVT, no upright pull	0.56	1.67	3.0×	unchanged

Training time excludes the per-epoch evaluation. One [GPU model] GPU per run. No upright pull: row (d) of Table VII. Inference: the same network, applied once to the upright image.

the other networks, and four rows of the component dissection are still pending.

VII. CONCLUSION

We asked which view should teach when a segmentation network is trained to agree with itself under flips. For binary labels, an identity answers it: a consistency pull toward a detached teacher view is the supervised pull, rescaled by one minus the ratio of the two views' errors. The pull therefore moves the student toward the label only where the teacher is closer to it. We turned this lag condition into FCT, which supervises an image and its two flips and lets a view teach another only where the label allows it, with a constant fixed by the decision threshold. FCT improved the unseen ETIS set for all three networks we tested, and the five-set mean Dice of Polyp-PVT and PraNet-V2 on every seed. It also made the predictions more flip-consistent and left inference unchanged. Future work will extend the identity to multi-class segmentation and to other transformations, such as rotations, and test FCT itself on the six PolypGen centres, where a label-aware variant already improves on every seed.

APPENDIX A
PER-SEED RESULTS

Table X lists the mDice of every run behind Tables I and II. FCT scores higher than vanilla with the same seed on the five-set mean for all twelve pairs, and on ETIS for eleven of them.

APPENDIX B
GAINS WITHOUT EPOCH SELECTION

Table XI repeats the comparison of Section V-C without epoch selection, with the evaluator that runs during training. At the last epoch, the ETIS gain of Polyp-PVT is the only entry of the table whose CI excludes zero; averaged over the last ten epochs, the gains on the five-set mean and on ETIS are positive for both networks, and their CIs exclude zero except for the five-set mean of Polyp-PVT. For PraNet-V2, the last epoch lies well past the peak of both arms (Fig. 5).

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TABLE X
PER-SEED MDICE (%) OF VANILLA AND FCT ON THE FIVE TEST SETS, BOTH NETWORKS.

	Polyp-PVT: vanilla / FCT						PraNet-V2: vanilla / FCT					
Seed	Kvasir	Clinic	Colon	300	ETIS	Mean	Kvasir	Clinic	Colon	300	ETIS	Mean
1	92.0/92.0	93.3/**93.5**	79.4/**80.4**	88.5/87.0	78.6/**79.8**	86.4/**86.5**	89.0/**89.7**	91.6/90.4	71.8/**72.0**	89.1/**89.7**	65.7/**71.4**	81.4/**82.6**
2	91.0/**92.7**	94.0/93.2	80.8/79.9	89.0/88.3	78.3/**81.3**	86.6/**87.1**	87.9/**89.2**	90.0/89.1	70.1/**73.8**	88.8/**89.5**	69.4/69.2	81.3/**82.2**
3	92.4/**92.8**	92.1/**93.7**	81.0/81.0	87.3/**89.9**	78.7/**80.1**	86.3/**87.5**	85.4/**89.4**	88.6/**90.2**	72.1/**74.4**	82.2/**87.6**	68.8/**69.8**	79.4/**82.3**
4	92.7/92.1	93.5/93.1	79.9/**81.3**	88.9/88.8	78.7/**79.6**	86.7/**87.0**	88.1/**90.7**	90.4/**91.1**	69.7/**72.6**	88.8/87.0	65.3/**73.7**	80.5/**83.0**
5	91.8/**93.0**	93.4/**93.5**	79.7/**79.9**	89.9/89.7	78.3/**80.1**	86.6/**87.2**	87.6/**89.4**	90.2/89.0	70.0/**73.7**	90.4/86.3	67.0/**69.5**	81.0/**81.6**
6	91.6/**92.4**	92.7/**93.1**	80.8/80.1	89.1/**89.4**	79.1/**80.8**	86.7/**87.2**	89.7/89.4	88.0/**89.2**	65.7/**71.0**	87.0/86.6	66.2/**70.5**	79.3/**81.3**

Bold: FCT above vanilla with the same seed (before rounding). Clinic: CVC-ClinicDB; Colon: CVC-ColonDB; 300: CVC-300.

TABLE XI
GAINS OF FCT WITHOUT EPOCH SELECTION: THE TRAINING-TIME EVALUATOR AT THE LAST EPOCH AND AVERAGED OVER THE LAST TEN EPOCHS (POINTS, 95% CI, SEEDS BETTER). BOLD: CIS THAT EXCLUDE ZERO.

Epochs	Five-set mean	ETIS
Polyp-PVT		
Last epoch	+0.55 [−0.11, 1.22], 4/6	**+2.60 [0.84, 4.36], 6/6**
Last 10 epochs	+0.42 [−0.03, 0.88], 5/6	**+1.52 [0.11, 2.93], 4/6**
PraNet-V2		
Last epoch	+0.49 [−1.17, 2.14], 4/6	−0.18 [−2.20, 1.84], 2/6
Last 10 epochs	**+2.41 [1.40, 3.43], 6/6**	**+3.71 [0.75, 6.68], 5/6**
ASGNet		
Last epoch	+0.13 [−0.91, 1.17], 3/6	+1.69 [−1.01, 4.39], 4/6
Last 10 epochs	+0.21 [−0.59, 1.01], 4/6	+1.99 [−0.26, 4.24], 4/6

Training-time evaluator: Dice of the min-max normalized probability map, per image, as in the official Polyp-PVT code. The six-seed mean of FCT is above that of vanilla after 81 of 99 epochs for Polyp-PVT, 28 of 29 epochs for PraNet-V2, 76 of 79 epochs for ASGNet.

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